

Evaluation Integrity in Machine Learning for Finance: A Pipeline-Aligned Systematization of Leakage, Backtest Overfitting, and Reproducibility Pitfalls — with a Practitioner Audit Checklist

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Abstract

Machine learning now sits at the center of empirical asset pricing, return forecasting, and quantitative strategy design, yet the headline numbers it produces are frequently inflated by avoidable evaluation errors rather than by genuine predictive signal. Two literatures have grown in parallel but rarely talk to each other: a general machine-learning literature on data leakage and the reproducibility crisis, and a quantitative-finance literature on backtest overfitting, multiple testing, and selection bias. This paper systematizes both into a single, pipeline-aligned account of evaluation integrity for financial machine learning. We map the canonical research-to-deployment pipeline — data sourcing, feature construction, sample splitting, model selection, backtesting, and live monitoring — to the evaluation-integrity failure modes that arise at each stage, the diagnostics that measure them (the optimism gap, the Deflated Sharpe Ratio, the Probability of Backtest Overfitting, purged and combinatorial cross-validation, and falsification audits), and the safeguards that contain them. From this mapping we distill two reusable artifacts: a stage-aligned taxonomy (Figure 1) and the Financial ML Evaluation-Integrity Checklist (FEIC), a fifteen-item pre-registration and audit instrument. We illustrate the safeguards with two open, offline-first reference implementations that quantify inflation against synthetic data with known ground truth. The contribution is integrative and methodological rather than a claim of tradable alpha; its purpose is to make evaluation assumptions, failure modes, and inflation directly measurable and reportable.

Keywords: *financial machine learning; data leakage; backtest overfitting; Deflated Sharpe Ratio; purged cross-validation; reproducibility; evaluation integrity; quantitative finance*

1. Introduction

The application of machine learning (ML) to finance has matured from a curiosity into a default methodology. Comparative studies of empirical asset pricing report large economic gains from tree ensembles and neural networks relative to linear baselines [1], and the approach has been replicated across the U.S., European, Chinese, international, REIT, and options markets [1, 14, 15, 16]. The central engineering question is no longer can ML forecast returns, but how much of any reported edge is real. A growing body of evidence suggests the honest answer is frequently: less than claimed.

The reasons are methodological rather than algorithmic. In the broader quantitative sciences, a systematic survey found data leakage in seventeen fields, affecting nearly 300 papers and, when corrected, often erasing the apparent advantage of complex models over decades-old logistic regression [2]. The same dynamics are sharper in finance, where serial dependence, non-stationarity, regime shifts, and a near-unlimited capacity to retry strategies make spurious discovery the norm rather than the exception. Foundational work shows that backtest performance is trivially inflated once a modest number of configurations is tried [4, 5], that most claimed factors in the cross-section of expected returns cannot survive a multiple-testing hurdle [9, 10], and that even when code and data are shared, fewer than half of published finance results reproduce exactly [11]. Recent papers go further, demonstrating that adaptive specification search manufactures statistically significant backtests even under a martingale-difference null [12] and that protocol choices — feature centering, execution timing — selectively inflate both predictive and trading metrics in controlled experiments [13].

Despite this evidence, practitioners lack a single map that connects the where (which pipeline stage), the why (the mechanism of inflation), the how-to-detect (the diagnostic), and the how-to-prevent (the safeguard). The general-ML leakage taxonomies [2, 3, 6] are not finance-aware: they do not address temporal embargoes, label horizons, or the deflation of performance statistics for multiple trials. The finance literature, in turn, treats backtest overfitting [4, 5, 7], cross-validation design [8], and reproducibility [11] as separate problems with separate tools. This fragmentation is itself a source of error: a team can adopt purged cross-validation yet still leak through global feature normalization, or report a Deflated Sharpe Ratio yet never embargo the label horizon.

This paper makes three contributions.

1. **A unifying construct — evaluation integrity.** We define evaluation integrity as the property that a reported performance estimate is an unbiased, multiplicity-aware, temporally honest predictor of out-of-sample behavior, and we argue that leakage and backtest overfitting are two faces of the same construct (Section 3).
2. **A pipeline-aligned taxonomy (Figure 1).** We map six canonical stages of the financial-ML lifecycle to their characteristic failure modes, the diagnostics that surface them, and the safeguards that contain them, consolidating evidence from both literatures into one table (Section 4, Table 1).
3. **A reusable audit artifact — the FEIC.** We distil the taxonomy into a fifteen-item Financial ML Evaluation-Integrity Checklist for pre-registration and review, and we demonstrate the underlying diagnostics in two open reference implementations that measure inflation against synthetic ground truth (Sections 5–6).

The work is a systematization of knowledge: it reports no new trading results and makes no claim of tradable alpha. Its aim is to give engineers, reviewers, and PhD researchers a common vocabulary and a concrete instrument for separating signal from artifact.

2. Background and Related Work

Machine learning in empirical asset pricing. The modern wave was defined by large-scale comparative studies establishing that nonlinear models capture predictor interactions missed by linear methods, with the dominant signals clustering around momentum, liquidity, and volatility [1]. Subsequent work extended the program internationally and into deep no-arbitrage formulations [16] and adjacent asset classes [14, 15]. These studies generate the economic incentive to deploy ML; they also generate the multiplicity that makes evaluation integrity hard.

Data leakage and reproducibility in ML-based science. Leakage — the contamination of the evaluation by information unavailable at prediction time — has been catalogued into structured taxonomies. Kapoor and Narayanan distinguish eight leakage types and propose model info sheets as a reporting remedy [2]; Sasse et al. give a complementary scenario-based overview [3]; Apicella et al. analyze how leakage propagates through ML and transfer-learning workflows [6]; and domain studies, such as connectome-based neuroimaging, quantify how feature-selection and subject-level leakage inflate performance, with small samples worsening the effect [17]. The shared lesson is that leakage is a pipeline property, not a single bad step.

Backtest overfitting, multiple testing, and selection bias. In finance the canonical results are sharper. Bailey, Borwein, López de Prado, and Zhu prove that minimal search suffices to produce an impressive backtest and name the practice backtest overfitting [4, 5], formalizing it through the Probability of Backtest Overfitting (PBO) via combinatorially symmetric cross-validation [5] and a Bayesian variant [18]. The Deflated Sharpe Ratio (DSR) corrects a reported Sharpe for the number of trials and for non-normal returns [19]. On the asset-pricing side, Harvey, Liu, and Zhu argue that a new factor should clear a t-statistic near 3.0 given decades of data mining [9], and Harvey and Liu calibrate false-discovery and missed-discovery rates jointly [10]. Most recently, Arian et al. compare out-of-sample protocols in a synthetic controlled environment and find combinatorial purged cross-validation (CPCV) dominant over walk-forward on PBO and DSR criteria [8]; Nikolopoulos introduces a falsification audit against zero-predictability reference classes [12]; and Zhang et al. construct a one-switch benchmark that isolates which protocol conventions cause inflation [13].

Reproducibility infrastructure. Pérignon et al. ran more than a thousand tests across research teams and found exact regeneration only about half the time, with reproducibility falling for complex code and tail results [11]. This situates evaluation integrity within a wider open-science agenda: a number can be both leakage-free and irreproducible.

The gap. No existing work places these strands on a single pipeline and pairs each failure mode with a diagnostic and a safeguard in one practitioner-facing instrument. Closing that gap is the purpose of this paper.

3. Evaluation Integrity: A Unifying Construct

We define **evaluation integrity** as the degree to which a reported performance estimate is an unbiased and honest predictor of true out-of-sample performance under three simultaneous conditions:

- **Temporal honesty** — every input used to produce a prediction for time t was knowable strictly before t , accounting for label horizons and information-release lags;
- **Sample hygiene** — no statistic estimated on test data (or on the full panel) leaks into training, including via preprocessing, feature selection, or normalization;
- **Multiplicity awareness** — the estimate is deflated for the number of strategy or model configurations that were searched, explicitly or implicitly, before it was selected.

Under this lens, leakage primarily violates temporal honesty and sample hygiene, while backtest overfitting primarily violates multiplicity awareness; but the two routinely co-occur, and a credible evaluation must satisfy all three. The construct also clarifies why a single fix is never sufficient: purged cross-validation restores temporal honesty but says nothing about multiplicity; the Deflated Sharpe Ratio restores multiplicity awareness but does not detect a leaky feature. The taxonomy in Section 4 is organized so that each stage is checked against whichever of the three conditions it most endangers.

4. A Pipeline-Aligned Taxonomy of Evaluation-Integrity Failures

Figure 1 organizes the financial-ML lifecycle into six stages and aligns each with its characteristic failure modes, detection diagnostics, and safeguards. Table 1 expands the same mapping with the mechanism of inflation and the integrity condition violated.



Figure 1. Stage-aligned taxonomy of evaluation-integrity failures in financial machine learning. Each stage lists its dominant failure mode, the diagnostic that surfaces it, and the safeguard that contains it.

The six stages are as follows.

Stage 1 — Data sourcing. Failure modes: survivorship bias (delisted names dropped from the panel), look-ahead from restated fundamentals or point-in-time errors, and vendor revisions silently overwriting history. These violate temporal honesty before a model is even trained. Safeguard: point-in-time databases with as-of timestamps, explicit delisting returns, and a frozen data snapshot hashed for reproducibility [11].

Stage 2 — Feature construction. Failure modes: global normalization or standardization computed over the full sample (a textbook leak [2, 3]), target leakage from features that encode the label, and centered or two-sided rolling statistics that peek forward. Zhang et al. show empirically that centered temporal features cause large, stable inflation [13]. Safeguard: fit all transforms inside the training fold only; use strictly causal (trailing) windows; audit each feature's information set.

Stage 3 — Splitting and cross-validation. Failure modes: random k-fold on serially dependent data, train/test adjacency without an embargo, and leakage across the label horizon. Safeguard: purged k-fold with an embargo around test windows, and combinatorial purged cross-validation (CPCV), which Arian et al. find superior to walk-forward on both PBO and DSR [8]. Walk-forward remains useful but exhibits weaker false-discovery control [8].

Stage 4 — Model selection and hyper-parameter optimization. Failure modes: tuning on the test set, nested-CV omission (the single most common optimism source [2]), and undocumented trial counts that hide multiplicity. Diagnostic: the optimism gap — the difference between the selection-time estimate and a held-out oracle — and PBO via combinatorially symmetric cross-validation [5]. Safeguard: nested cross-validation, pre-registered search spaces, and a trial counter that feeds the deflation in Stage 5.

Stage 5 — Backtesting and performance reporting. Failure modes: selection bias from reporting only the best run, non-normal-return inflation of the Sharpe ratio, and ignored transaction costs and market impact. Diagnostic: the Deflated Sharpe Ratio [19], the t-statistic hurdle near 3.0 for new factors [9], and falsification against zero-predictability placebos [12]. Safeguard: report DSR with the trial count, include realistic costs, and pass a placebo/falsification audit before claiming a result.

Stage 6 — Deployment and monitoring. Failure modes: regime shift and concept drift, decay of crowded signals, and execution slippage absent from the backtest. Safeguard: live paper-trading with pre-committed stop rules, drift monitors, and periodic re-deflation as new trials accumulate.

Table 1. *Evaluation-integrity failures mapped to mechanism, diagnostic, safeguard, and the integrity condition violated.*

Stage	Failure mode	Mechanism of inflation	Diagnostic / metric	Safeguard	Condition violated
1. Data sourcing	Survivorship & look-ahead bias	Winners over-represented; restated data	Point-in-time audit; snapshot hash	As-of timestamps; delisting returns; frozen, hashed data	Temporal honesty
2. Feature construction	Global normalization; centered windows	Test-set statistics enter training	Per-feature information-set audit	Fold-internal transforms; causal windows	Sample hygiene
3. Splitting & CV	Random k-fold; no embargo	Adjacent dependent samples shared	PBO; CPCV comparison [5, 8]	Purged k-fold + embargo; CPCV	Temporal honesty
4. Model selection / HPO	Tuning on test; no nested CV	Selection optimism	Optimism gap; nested-CV delta	Nested CV; pre-registered grid; trial counter	Sample hygiene + multiplicity
5. Backtest & reporting	Best-run reporting; cost omission	Multiple-trial selection bias	Deflated Sharpe Ratio [19]; $t \approx 3$ hurdle [9]; falsification [12]	DSR with trial count; realistic costs; placebo test	Multiplicity awareness
6. Deployment	Regime shift; signal decay	Distribution shift post-fit	Live drift monitors; paper-trade gap	Pre-committed rules; re-deflation; monitoring	Temporal honesty (ongoing)

5. The Financial ML Evaluation-Integrity Checklist (FEIC)

The taxonomy condenses into a fifteen-item instrument intended to be filled out before a result is reported, in the spirit of the model info sheets proposed for ML-based science [2] and the reproducibility guidelines emerging in finance [11]. Each item names the stage it guards.

1. **(S1)** Is the dataset point-in-time, with as-of timestamps and delisting returns included?
2. **(S1)** Is the exact data snapshot frozen and hashed, with vendor version recorded?
3. **(S2)** Are all transforms (scaling, imputation, selection) fit inside the training fold only?
4. **(S2)** Are all rolling features strictly causal (trailing), never centered or two-sided?
5. **(S2)** Has each feature been audited so that its information set predates the label?
6. **(S3)** Does cross-validation purge and embargo around the label horizon?
7. **(S3)** Is CPCV or purged walk-forward used rather than random k-fold?
8. **(S4)** Is hyper-parameter search wrapped in nested cross-validation?
9. **(S4)** Is the search space pre-registered and the total trial count logged?
10. **(S4)** Is the optimism gap (selection estimate minus held-out oracle) reported?
11. **(S5)** Is the Sharpe ratio deflated for the trial count and return non-normality (DSR)?
12. **(S5)** Are realistic transaction costs and market impact included?
13. **(S5)** Does the strategy survive a falsification/placebo audit against a zero-predictability null?
14. **(S6)** Is there a pre-committed live paper-trading window with stop rules before capital allocation?
15. **(All)** Is code and a reproduction script released so a third party can regenerate the headline numbers?

The checklist is deliberately tool-agnostic: items 6–11 can be satisfied with widely used open libraries, and items 1, 2, and 15 align the result with computational-reproducibility norms [11].

6. Illustrative Reference Implementations

To keep the framework concrete and reproducible, we point to two open, offline-first benchmark tools that operationalize the Stage-4 and Stage-5 diagnostics. Both synthesize data with a known ground truth, then measure how far a given protocol's reported performance departs from it; neither makes any claim about real markets, and no live-trading results are reported here.

- **MLForge** (github.com/ranafaraz; demo at mlforge.dexdevs.com) contrasts leaky, single-split, and nested cross-validation against a held-out oracle on synthetic data, and reports the resulting optimism gap. It instantiates checklist items 8–10 and the Stage-4 diagnostic in Table 1.

- **FinSight** (github.com/ranafaraz; demo at finsight.dexdevs.com) implements an overfitting-aware backtest with purged walk-forward cross-validation and the Deflated Sharpe Ratio, exposing how trial count and return non-normality deflate an apparent edge. It instantiates checklist items 7, 11, and 12 and the Stage-5 diagnostics.

These tools are illustrative artifacts that make the safeguards executable; they are not evidence of predictive performance and are reported here only to demonstrate that each checklist item has a working, inspectable reference. The design — synthetic ground truth plus an ablation of which method recovers it — mirrors the synthetic-controlled-environment methodology used to compare cross-validation protocols in the literature [8] and the one-switch isolation of protocol effects proposed by Zhang et al. [13].

7. Discussion

Conceptual versus empirical. Every claim in this paper is either (a) a synthesis of results reported in the cited literature, attributed inline, or (b) a definitional or organizational proposal (the evaluation-integrity construct, the taxonomy, and the FEIC). No new empirical trading result is asserted. Where we describe the reference implementations, we describe what they measure (optimism gap, Deflated Sharpe Ratio against synthetic ground truth), not any market outcome.

Why a unified instrument matters. The recurring pattern in the evidence is that teams fix one integrity condition and assume the others follow. A purged-CV pipeline with global normalization still leaks; a DSR-reported strategy with an unembargoed label horizon is still temporally dishonest. The value of a single stage-aligned checklist is that it forces all three conditions — temporal honesty, sample hygiene, multiplicity awareness — to be checked together, at the stage where each is most at risk.

Relationship to reproducibility. Evaluation integrity and computational reproducibility are complementary but distinct. A result can be reproducible (the code regenerates the number) yet lack integrity (the number is leaked or overfit), and vice versa. Item 15 of the FEIC bridges the two by requiring a third-party-runnable reproduction script, aligning the instrument with the reproducibility guidelines emerging in finance [11].

8. Limitations

This is a conceptual systematization and inherits the usual limits of that genre. First, the taxonomy is a simplification: real pipelines have feedback loops (deployment feeds new data into sourcing) that a six-stage linear diagram understates. Second, the checklist is necessarily generic; specific asset classes (high-frequency, options, credit) carry idiosyncratic leakage channels not enumerated here. Third, we synthesize a literature that is itself still consolidating — recent benchmarks for decision-time leakage [13] and falsification audits [12] are new and not yet widely replicated, so some recommended diagnostics rest on a thin empirical base, which we flag rather than overstate. Fourth, the reference implementations demonstrate diagnostics on synthetic data; transfer to a particular institution's data and governance context requires the user's own validation. None of

these limits undercuts the core proposal — that leakage and backtest overfitting are one construct best audited stage by stage — but each marks where future empirical work is needed.

9. Conclusion and Future Work

Machine learning has given finance enormous modeling power and an equally enormous capacity to fool itself. The literature has answered the diagnosis problem in pieces — leakage taxonomies on one side, backtest-overfitting statistics on the other — but practitioners have lacked a single map from pipeline stage to failure mode to diagnostic to safeguard. This paper supplies that map: a unifying construct (evaluation integrity), a stage-aligned taxonomy (Figure 1, Table 1), and a fifteen-item audit checklist (the FEIC), each grounded in current evidence and paired with open reference implementations. Three directions follow. First, the FEIC should be validated empirically — does adopting it measurably reduce the optimism gap across teams, as the reproducibility study design [11] would allow one to test? Second, asset-class-specific extensions are needed for high-frequency and derivatives settings. Third, as agentic and large-language-model pipelines enter quantitative finance, decision-time and temporal-knowledge leakage [13] will require new diagnostics that the present taxonomy can host but does not yet fully specify. The immediate, actionable contribution is simpler: a number reported from a financial-ML pipeline should travel with its FEIC, so that a reader can see not just the result but the integrity of the evaluation that produced it.

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